

Problem 3.21

Suppose that $i_n = 0.01n$ for $n \geq 1$. Show that

$$I_n = 0.01n(1.01)(1.02) \cdots [1 + 0.01(n-1)]A(0).$$

Solution

We are given

$$i_n = 0.01n.$$

For example,

$$i_1 = 0.01, \quad i_2 = 0.01 \cdot 2 = 0.02, \quad i_3 = 0.01 \cdot 3 = 0.03,$$

and in general,

$$i_{n-1} = 0.01(n-1).$$

Also,

$$I_n = A(n) - A(n-1),$$

and

$$i_n = \frac{I_n}{A(n-1)}.$$

Therefore,

$$I_n = i_n A(n-1) = 0.01n A(n-1).$$

Now use the accumulation recursion:

$$A(1) = A(0)(1 + i_1),$$

$$A(2) = A(1)(1 + i_2),$$

$$\vdots$$

$$A(n-1) = A(n-2)(1 + i_{n-1}).$$

Substituting repeatedly, we get

$$A(n-1) = A(0)(1 + i_1)(1 + i_2)(1 + i_3) \cdots (1 + i_{n-1}).$$

Since

$$i_k = 0.01k,$$

this becomes

$$A(n-1) = A(0)(1.01)(1.02) \cdots (1 + 0.01(n-1)).$$

Hence,

$$I_n = (0.01n) \left[A(0)(1.01)(1.02) \cdots (1 + 0.01(n-1)) \right].$$

Therefore,

$$\boxed{I_n = 0.01n(1.01)(1.02) \cdots [1 + 0.01(n-1)]A(0).}$$